Project thesis

Toxicity prediction of plastic chemicals

Amelie Hübner

supervised by

Prof. Dr. Franziska Lederer [1]
Dr. Sebastian Starke [2]

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[1]Helmholtz-Zentrum Dresden-Rossendorf, Helmholtz Institute Freiberg for Resource Technology, Freiberg, Germany

[2]Helmholtz-Zentrum Dresden-Rossendorf, Department of Information Services and Computing, Dresden, Germany

Contents

1 Introduction

The impact and scale of plastic production and consumption on the environment is steadily increasing. In 2020, the world-wide cumulative primary plastic production surpassed 10 000 million tonnes. Only around 10% of global plastic waste has ever been recycled. Of the remaining 90%, 14% have been incinerated, producing greenhouse gas emissions and releasing gaseous pollutants in the process. The remaining 76% stay intact, leaching toxic chemicals and microplastics into the environment (Walker 2025).

A multitude of polymer groups are used in industrial settings, which can be differentiated based on their polymer backbone. The largest groups in commodity plastics are polyethylene (PE; 36%), polypropylene (PP; 21%) and polyvinylchloride (PVC; 12%), followed by polyethylene terephthalate (PET), polyurethane (PUR) and polystyrene (PS) (less than 10% each). In total, these groups add up to 92% of all plastics ever produced (Geyer et al. 2017).

The wide variety and usability of plastic products stems from the immense range of chemicals that are added during the manufacturing process for different reasons. Without additional chemicals, polymers are not broadly usable or durable (Wagner et al. 2025). Additives aid in manipulating the plastic products properties, such as plasticizers, which are widely used to make the endproduct more flexible, or colorants, which give the endproduct its color (Hahladakis et al. 2018). Processing aids, such as catalysts and solvents, help during the production and processing of polymers. Non-intentionally added substances (NIAS) can be categorized into side products, breakdown products and contaminants. The extent to which manufacturers are aware of the presence of these substances in the end product varies. It is evident that the knowledge and classification of NIAS is, in general, limited, due to a lack of both information and toxicity testing. This is partly due to complex structures such as oligomeres being present (Geueke 2018).

Plastic chemicals are used to give the end product specialist functions, while often being the cause of the end product having toxic properties. Humans are at risk of plastic chemical exposure during every step of the products life cycle either directly (e.g., plastic straws, plastic childrens toys) or indirectly (e.g., ingestion of food packaged in plastic, inhalation of plastic chemicals from air). Plastic chemicals have been linked to many adverse health effects, such as nonylphenol (a plasticizer) and bisphenol A (used as a starting material for polymer synthesis) being endocrine disruptors. Plastic additives bonds to polymers are weak and therefore pose risks of them easily being released into the environment at any stage during the plastic life cycle.

The reusability and recycling capacity of the plastic product is influenced by the chemicals as well, as high quality recycling products can often only be produced from well-separated clean recyclates, that is, recyclates free of most plastic chemicals. Additionally, chlorinated plastic chemicals can lead to corrosion on recycling machinery. Even though of all these problems, plastic chemicals remain widely unregulated and undocumented and often lack proper hazard information (Wagner et al. 2025).

PUR is a highly diverse group of synthetic thermoplastic polymers created via polycondensation reactions between isocyanates and compounds containing hydroxyl. The wide range of possible reactants forming polyurethanes leads to their usage in a multitude of industries as sealants, coatings, elastomers, and foams. Recycling them is of importance considering the ongoing depletion of fossil fuel reserves (Rossignolo et al. 2024).

Plastic products are primarily recycled mechanically or chemically by depolymerization. Mechanical recycling focuses on thermoplastics such as polyethylene (PE) and polyethylene terephthalate (PET), which consequently are the most recycled plastics.

The following report predicts the toxicity of plastic chemicals present in the dataset released by Wagner et al. 2025 by training a machine learning model using XGBoost (Chen and Guestrin 2016). They present 16 325 known plastic chemicals and their properties compiled from multiple open databases. Only around 6% of them are internationally regulated, and more than 10 000 are missing any hazard information at all. This motivates a structure-based toxicity prediction approach. A significant challenge in the implementation of this approach is the significant class imbalance of the available hazard information, which is addressed via a combination of resampling and a custom loss objective.

2 Methods

2.1 Dataset

The PlastChem database is the largest publicly available plastic chemical database to date. It was built to address the scarcity and scattering of chemical data on plastic chemicals, a problem faced by policymakers and researchers in the plastic sector, and compiled and identified 17 932 entries which are present in plastic products. The database was created through a systematic collection and harmonization of information from scientific and regulatory sources. In addition to identifiers, it contains information about chemical properties, regulation, hazard information, structural groups, production and functions across multiple plastic subgroups. The authors identified chemicals of concern based on their hazard information, which was determined on four criteria: persistence, bioaccumulation, mobility and toxicity. These four criteria were transferred to a single hazard score. For more information on the hazard scores, see Supplementary information of Monclús et al. 2025 and figure S1.

Across the entire database, around 25% of all listed chemicals have been assigned a hazard score. The 10 largest polymer groups, that is, the 10 groups with the most chemicals assigned to them, each contain at least 400 chemicals of concern. This work uses the PlastChem database (version 1.1, Wagner et al. 2025) as the source for both molecular structures and hazard labels for training a classification model.

For the purpose of this project, only compounds meeting both the following criteria were used for model training and evaluation:

1. a canonical SMILES[3] string was present, allowing computation of Morgan molecular fingerprints[4] as a model input, and
2. the hazard score label was present, allowing its direct usage as a toxicity label.

Entries failing either criterion were excluded, as neither a valid molecular representation nor a usable target label could be constructed for them. Compounds for which fingerprint generation failed (e.g. due to invalid SMILES) were removed as well. This resulted in 4 748 compounds that were deemed usable for the model training. Finally, hazard scores were binarized into hazardous and not hazardous, by merging the intermediate and high-hazard categories into a single class.

The resulting dataset is strongly imbalanced toward the hazardous class across all polymer types. In total, the hazardous class represents 97% of the data. Figure 1 shows the imbalance per polymer type. Across all ten most abundant polymers, the hazardous class makes up between 95% and 98% of compounds, with PET and PP being associated with the largest absolute number of chemicals of concern and PC (polycarbonate) and ABS (acrylonitrile butadiene styrene) with the lowest amount. As this is only a representation of known plastic chemicals, these statements cannot be transferred to real world observations.

[3] Simplified Molecular Input Line Entry System; a representation of molecular structures as ASCII-strings

[4] binary encoding of molecular structures

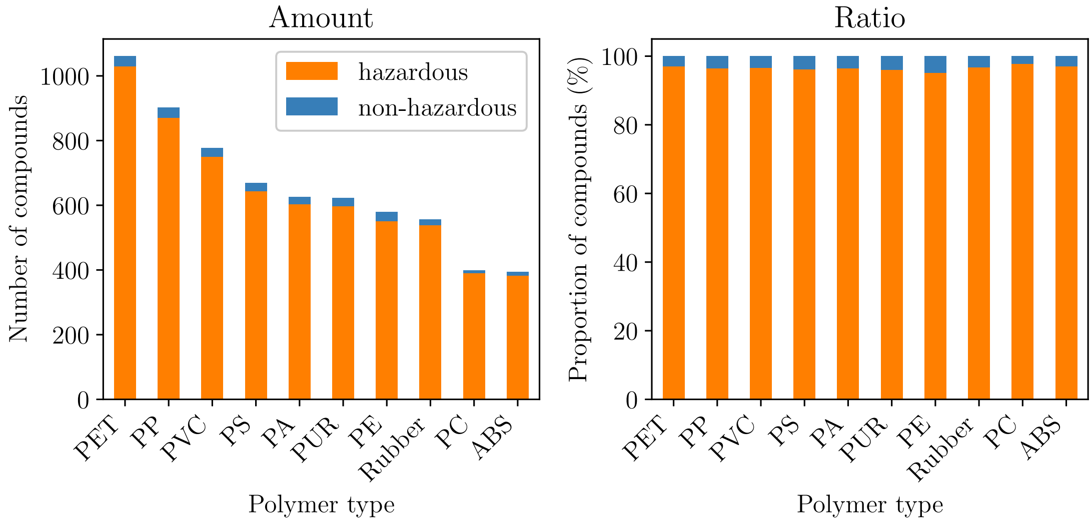

Figure 1: Amount and ratio of hazardous and non-hazardous compounds in the dataset displayed by polymer type for the ten most abundant polymer types present.

This pronounced imbalance is the primary reason resampling was incorporated into the training pipeline described in section 2.2, and why balanced accuracy rather than raw accuracy was used as the model selection criterion during hyperparameter search.

Another complexity the dataset presents is its high dimensionality. This is invariably the case when dealing with chemical data represented as high-dimensional sparse fingerprints. Figure 2 visualizes this using a two-component PCA projection of the Morgan fingerprints, in which the two components together capture just 8.56% of total variance.

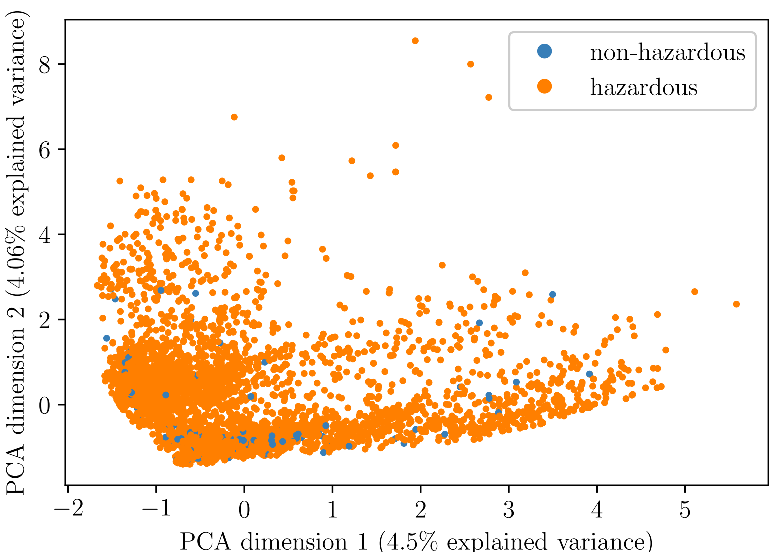

Figure 2: Two-dimensional principal component analysis of dataset represented as molecular fingerprints, explaining 8.56% of total variance.

Hazardous compounds (blue) are distributed broadly across the whole projected space, while non-hazardous compounds (orange) are comparatively rare, with no clear linear separation visible in the first two principal components. XGBoost's ability to model non-linear relationships is well suited to this this lack of linear separability in a low-dimensional projection.

2.2 Model training

The aim of this project is to build a binary classifier, which, given the molecular structure of a plastic associated chemical encoded as a fingerprint, predicts whether the compound is hazardous or not hazardous. As described in section 2.1, the dataset is highly imbalanced and not very well separable. In order to build a reliable classifier, the decision was made to binarize the hazard labels, as multi class classifiers are even more susceptible to imbalance.

The dataset (the remaining 4 748 compounds with a valid SMILES-derived fingerprint and a hazard-label, see section 2.1) was split into three disjoint sets, with 70% of the data used as the exploration set for a 5-fold cross-validated hyperparameter search, and the remaining 30% split in half into test and validation set (15% each). While the validation set was used for design improvements, the results in section 3 are all reported on the independent test set. All splits were stratified on the target hazard label in order to preserve class the previously described class imbalance, and a fixed random seed was used throughout for reproducibility, as implemented in the scikit-learn python package (Pedregosa et al. 2011).

To prevent data leakage, preprocessing and classifier training were combined into a single pipeline. A grid search 5-fold cross-validation approach was taken to determine the best parameter values to fit the training data on the classifier. First, a Principal Component Analysis (PCA) was applied to transfer the 2048-bit large binary fingerprints into a continous lower-dimensional representation. Then, the pipeline applies the Synthetic Minority Over-sampling TEchnique algorithm (SMOTE) combined with Edited Nearest Neighbors (ENN) undersampling. This approach oversamples the minority class by interpolating between existing samples in order to balance the class ratio. Subsequently, the sample space was cleaned using ENN. The final classifier is an XGBoost classifier employing a histogram-based tree method.

Morgan fingerprints are high-dimensional, sparse binary vectors, in which distance-based methods such as ENN and SMOTE are known to perform poorly due to the curse of dimensionality (Altman and Krzywinski 2018) especially considering the dataset size. Therefore, PCA was applied first to obtain a lower-dimensional, dense representation in which relationships between neighboring molecules are more meaningful. The optimal number of components for the PCA was determined during the grid search.

SMOTE (Chawla et al. 2002) was selected instead of simple random oversampling or class weighting because it generates synthetic minority-class samples by interpolation rather than duplication, which reduces the risk of overfitting the model to repeated examples. As a way to counteract the synthetic points in ambiguous or overlapping regions of the feature space introduced by SMOTE, it was followed by an ENN (Wilson 1972) step step to remove noisy majority-class samples near the decision boundary. Combining SMOTE and ENN follows the recommendation for a combined resampling by Batista et al. 2005. The combined scikit-learn implementation SMOTEENN (Lemaître et al. 2016) has a parameter to set the preferred ratio of minority class over majority class samples after resampling, which was optimized in the grid search.

XGBoost (Chen and Guestrin 2016) was used as the classifier due to its strong benchmark performance on tabular and molecular fingerprint data (see Ding et al. 2021 for an example). It employs a gradient boosting approach in which decision trees are added iteratively to improve the ensemble's predictions on the training data. The final prediction for each sample is determined based on the combined predictions of all decision trees in the ensemble. The XGBoost framework is especially useful for data in which one of the challenges is to prevent overfitting, as in the herein presented severely imbalanced data. For the XGBoost classifier, the tree method "hist" was chosen because of its performance advantages. Further, the maximum delta step parameter is set to 3 in order to restrict the weight of each tree, another measure taken to prevent overfitting on the majority class.

Additionally, a custom focal loss objective implemented by Wang et al. was adapted for the scikit-learn API and used in the classifier. Focal loss was proposed by Lin et al. with the goal to address binary class imbalance by reducing the loss allocated to well-classified samples. The focusing parameter was set to 2, as the authors described this value to work best. Because of time and implementation constraints, this parameter was not optimized in the grid search.

Finally, the four following parameters were optimized during the grid search:

- the maximum depth of each tree, restricting the size of each decision tree
- the learning rate of the classifier, restricting the impact of each new tree on the prediction
- the number of estimators, that is, decision trees forming the ensemble, and
- the subsample ratio of columns for each level, a further measure to prevent overfitting by reducing the number of features seen by the classifier during each tree level.

The results for each parameter that was optimized in the aforementioned grid search are displayed in section 3.

2.3 Computing environment

The model training was conducted on a compute node of the Friedrich-Schiller-University high performance computing cluster 'Draco'. The node is equipped with two Intel Xeon Gold 6326 CPUs with 2.90 GHz. The system has a total of 32 physical CPU cores or 64 logical CPUs. For the training, 32 CPU cores were allocated with 32 GiB of memory. All model training was executed using Python 3.14.6. The primary software libraries used for data processing and machine learning included NumPy 2.5.0, pandas 3.0.3, SciPy 1.18.0, scikit-learn 1.9.0, XGBoost 3.3.0 and imbalanced-learn 0.14.2.

3 Results

3.1 Grid search

An iterative grid search was performed, progressively narrowing the search range around promising values chosen from initial exploration runs (Table 1). The final configuration selected "n_components=130", "sampling_strategy=0.4", "max_depth=5", "learning_rate=0.01", "n_estimators=200", and "colsample_bylevel=0.75", achieving a cross-validated balanced accuracy of 0.615 (see section 3.3 and figure 4) over 5 folds. Notably, both max_depth and sampling_strategy were selected at the boundary of their respective search ranges. The search grid for the sampling_strategy parameter was not extended further, in order to avoid the introduction of additional noise.

Pipeline step	Parameter	Explanation	Grid values
PCA	"n_components"	No. of principal components	100, <u>130</u>, 160
SMOTEENN	"sampling_strategy"	desired class ratio	0.2, 0.3, <u>0.4</u>
XGBoost	"max_depth"	max. depth of trees	<u>5</u>, 7, 9
XGBoost	"learning_rate"	impact of new trees	0.001, <u>0.01</u>, 0.1
XGBoost	"n_estimators"	No. of decision trees	100, <u>200</u>, 300
XGBoost	"colsample_bylevel"	feature reduction	0.5, <u>0.75</u>, 1.0

Table 1: Results of iterative grid search, with chosen parameters underlined.

The grid search ran for a total CPU time of around 2 days, with a wall time of around 2 hours.

3.2 Data processing

As described in table 1, the grid search for the optimal number of principal components resulted in 130. This number of components then explained a total variance of 51.41%.

Figure 3 illustrates the effect of the applied resampling strategy on the previously described class imbalance. Before resampling, the training set consisted of 3 323 total samples, with 109 labelled as non-hazardous, while the remaining 3 214 were assigned to the hazardous group. After resampling, the training set consisted of a total of 3 865 samples, 1 252 of which are labelled as non-hazardous, and 2 613 as hazardous. The imbalance between the classes is considerably reduced by the described approach. The resulting resampled dataset was subsequently used for the model training.

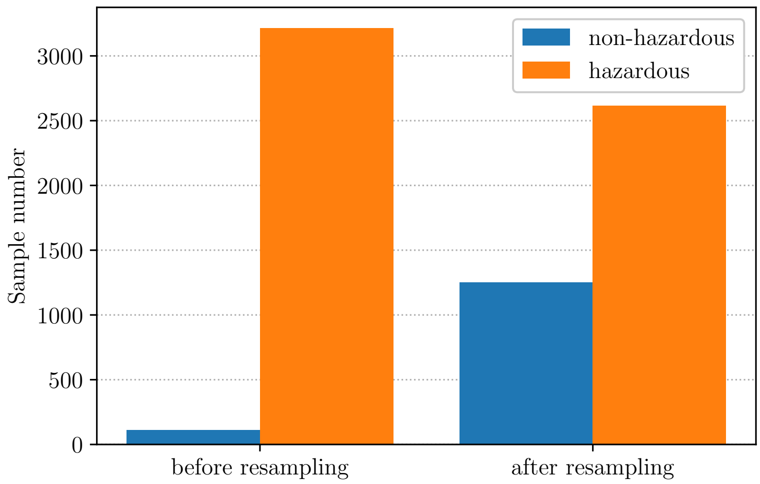

Figure 3: Effect of SMOTEENN resampling on the class distribution of the training set.

3.3 Performance

Figure 4 displays the balanced accuracy across the five cross-validation folds of the resulting optimized model. Notably, the performance varied between folds, ranging from around 0.517 to around 0.708. The mean balanced accuracy as indicated by the black line reached 0.615. On the independent test set, a slightly higher balanced accuracy of 0.639 was achieved. Overall, the results show substantial variation between the individual cross-validation folds, while the test set achieved a similar balanced accuracy score to the mean cross-validated score.

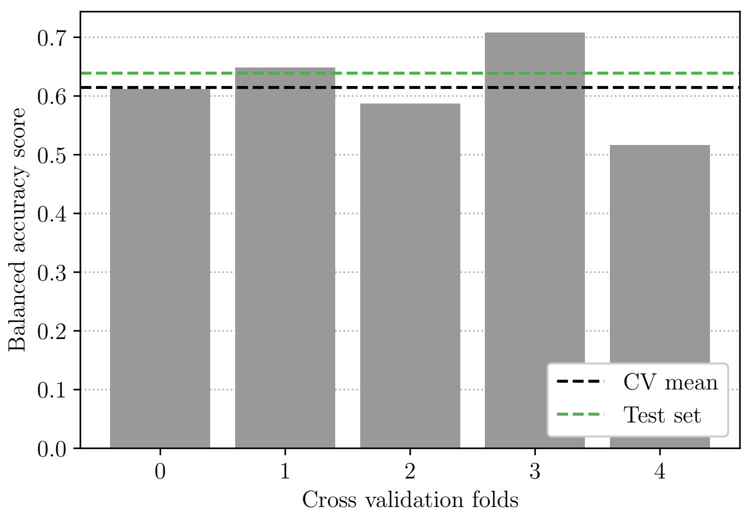

Figure 4: Balanced accuracy across the five cross-validation folds. Dashed black line indicates mean balanced accuracy across all folds. Dashed green line indicates balanced accuracy of test set.

The model reached a balanced accuracy score of 0.639 and a ROC-AUC score of 0.763 on the test set. As seen in table 2, the model achieved vastly different performances on the two classes. For the majority class, it performs well, achieving a precision of 0.98 with a recall of 0.9 and a F1 of 0.94. On the minority class, it severely underperforms, achieving a precision of 0.12, a recall of 0.38 and a F1 score of 0.18. The support for the minority class is 24, indicating that every classified compound has an effect of around 4% on the reported recall and around 1% on the precision. The model appears to be strongly optimized towards the majority class. While the results presented below indicate its reliable detection of hazardous compounds, the performance for the non-hazardous class is notably low and appears to be a significant limitation of the model, further suggesting that minority-class predictions should be interpreted with caution. These measures are additionally supported by figure S2.

Class	Precision	Recall	F1-Score	Support
non-hazardous	0.12	0.38	0.18	24
hazardous	0.98	0.90	0.94	689

Table 2: Classification performance on the test set.

Figure 5 shows precision-recall curves for both classes on the independent test set using the final model. The hazardous class achieves an average precision (AP) of 0.99, while the minority class reaches an AP of 0.02, below its class occurrence of approximately 0.034. Section 4 discusses the difference between the two curves and how it impacts the model reliability.

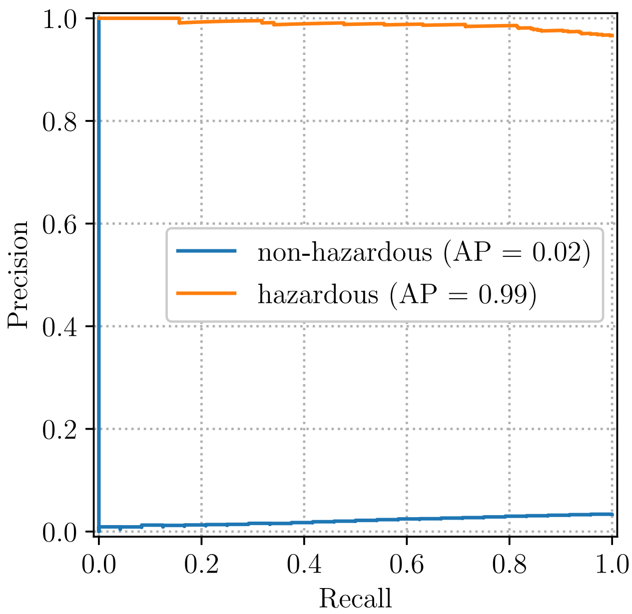

Figure 5: Precision-recall curves for both classes on the independent test set (n=713, of which 24 labelled as non-hazardous).

A comparison of the final model against a most-frequent dummy classifier (always predicts the majority class) can be seen in figure 6. Notably, the model outperforms the dummy classifier in this measure, indicating predictive performance above chance. The discrepancy between the metrics displayed in 5 and figure 6 is further examined in section 4.

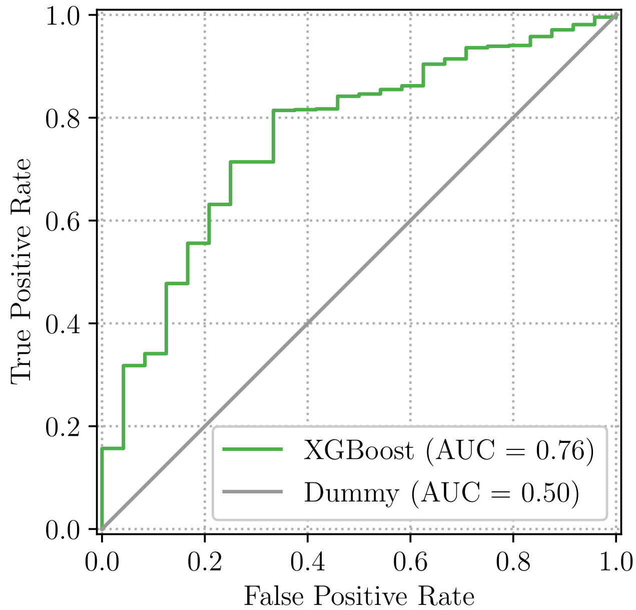

Figure 6: ROC curve for the final XGBoost model on the independet test set, compared against a dummy baseline.

See figure S3 and figure S2 for confusion matrices of a dummy classifier and the trained model's performance on the test set.

4 Discussion

4.1 Dataset limitations

The main limitation of this project thesis is the limited data availability. Imbalancedness of the previously described degree introduces overfitting to the majority class, which the results in section 3.3 illustrate. The model is unable to learn the structure of non-hazardous compounds given their limited availability in the training data. Resampling strategies such as SMOTEENN improve the models performance but have to be viewed critically, as it cannot be verified whether synthetic interpolations compound to naturally occurring (unknown) compounds. Representing the compounds using morgan fingerprints is limited to the fingerprint size. Different fingerprinting techniques have different advantages and may be more useful for harnessing the toxicity signal.

More data availability would increase the predictive power, but the plastic industry has no incentive to release data that could potentially harm their sales. The scarcity present in the dataset is a property of how the underlying data was generated. As discussed in section 2.1, more than 10 000 of the 16 325 plastic chemicals present in the PlastChem database lack any hazard information at all. Additionally, there is a bias in the data that is already released: the more toxic a compound is, the more likely its toxicity will be investigated. As a result, the present labels reflect regulatory and industrial testing priorities instead of the true frequency of hazard present in plastic chemicals. This is related to a broader issue already raised in the Introduction: plastic product manufacturers are not required to disclose toxicological information for plastic chemicals (Geueke 2018), so the class imbalance observed is symptomatic of the regulatory shortcomings rather than a sampling issue.

4.2 Model performance metrics

The data imbalance also has a significant impact on the reliability of the model performance metrics. The per-class performance is asymmetric: the model correctly identifies 90% of hazardous compounds, while simultaneously only 12% of the compounds it predicts as non-hazardous are correctly identified. As the minority class consists of only 24 samples, every sample shifts the recall by around 4%. The performance metrics for the minority class should therefore be interpreted as noisy estimates rather than robust measures. This is also underlined by the cross validation performance across the five folds displayed in figure 4.

Additionally, figure 5 and figure 6 differ in their conveyed message due to the imbalance. The precision is reported at a threshold of 0.5 while the average precision (AP) reported in 5 is averaged over the precision at all classification thresholds, weighted by the recall gained at each. As the precision-recall curve for the non-hazardous class lies close to its class occurrence in all areas except the baseline near the default threshold, the resulting AP (0.02) is lower than the precision 0.12 reported at threshold 0.5. The threshold 0.5 shows one rather favorable point compared to the otherwise weak curve. For the minority class, the model shows weak precision across thresholds. While the performance seems overtly optimistic in the ROC-AUC curve, it falls short in the precision recall metric for the minority class. Partially, the reason for this divergence lies in the fact that the model was trained on SMOTEENN-rebalanced data but evaluated at a threshold not tuned for the imbalanced test set distribution.

Threshold rebalancing was considered but was omitted from this work due to time constraints. Considering the scale of the precision gap, its effect may also be rather small.

4.3 Generalization limitations

As novel plastic chemical compounds are synthesized linearly by adding active groups to already existing compounds, the chemical space of known compounds can be separated into clusters. This introduces bias into models trained using random splitting, where those clusters will be evenly distributed, and allows the models to memorize compounds and subsequently predict their toxicity based on the neighbouring compounds (Riley 2019), resulting in overtly optimistic predictive performance (Guo et al. 2024). The chemical space of known molecules is complex, indicating that the chemical space of unknown compounds is even more complex.

The finding that a comparatively basic pipeline achieves discriminative performance above chance demonstrates that structure based approaches have potential to reliably harness toxicity signals. Toxicity predicition from chemical structures is an active, substantially larger research area where substantial theoretical research is still conducted. However, building a de novo architecture better at detecting the chemical signals and training a model with increased complexity is beyond the scope of this work.

4.4 Outlook

Several steps are obvious following the approach of this work. First, the random split does not transfer to truly novel compounds, therefore more realistic ways of data splits need to be investigated for this problem, such as scaffold splits (which come with their own limitations, see Kretschmer et al. 2025) or maximum common edge subgraph splits (Raymond et al. 2002), which give more realistic estimates of performance on compounds dissimilar to those seen in training. Second, density-based clustering methods (e.g., DBSCAN) could be better suited to the sparse, non-linear structure of binary fingerprints rather than a variance-based linear method such as PCA. Incorporating the class imbalance into the model directly as a prior rather than through resampling could be evaluated against the here employed SMOTEENN approach. Additionally, other methods of fingerprint generation should be investigated, see Huber and Pollmann 2026 for a comprehensive review on possible representations. Finally, alternative model architectures, such as graph neural networks operating directly on the molecular structure should be considered (see Atz et al. 2021 for an extensive review).

This work presents a primary investigation into applicability of the PlastChem database for structure-based hazard prediction. It underlines the inherent complexity of chemical machine learning and illustrates that toxicity predicition from molecular structures is a multidimensional problem, restrained by data availability and representation choices as well as model architecture. The scope of this work is too limited to sufficiently solve this problem, but rather serves as a stepping stone for further, more extensive work. Rather than an exact substitute for an experimental hazard assessment, the model presented in this work could serve as a primitive filter for triaging novel chemicals of concern.

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S1 Code availability

All code used for data processing, training, model implementation and visualization can be found under the project thesis github.

S2 Additional Figure

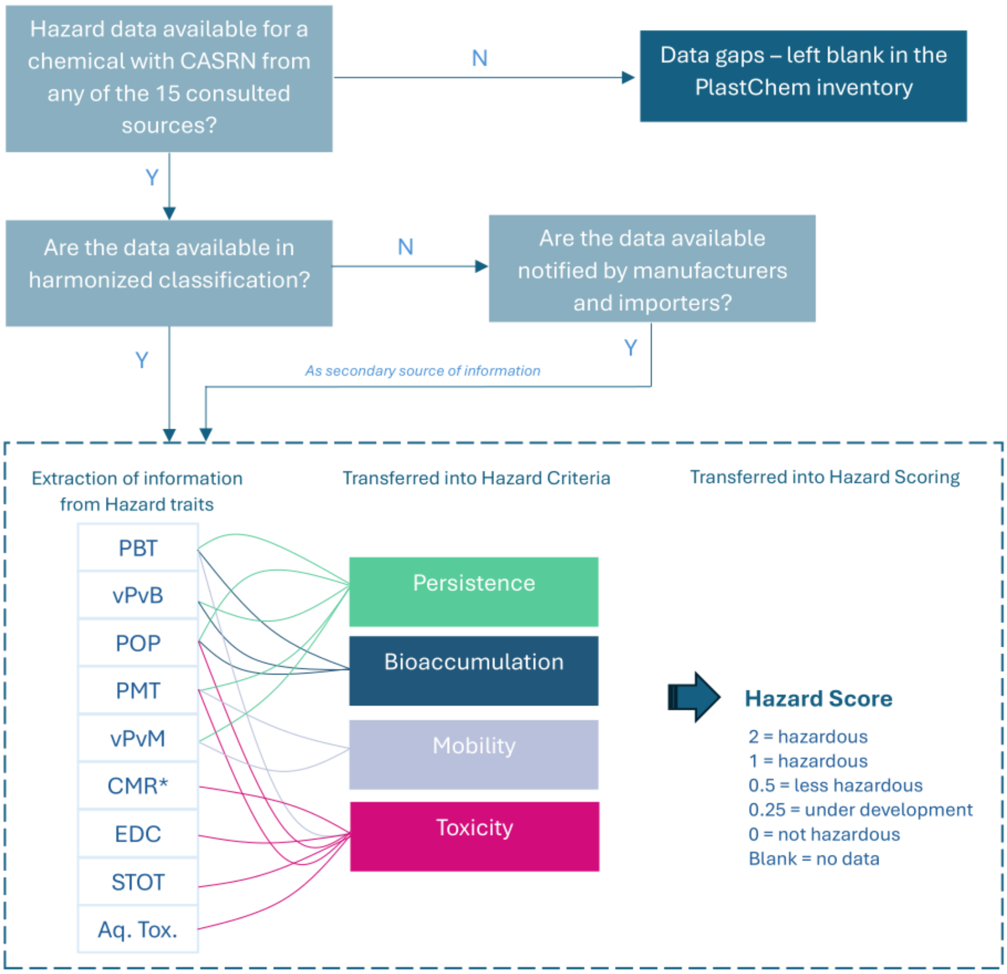

Figure S1: "Workflow to assign hazard scores. PBT refers to persistent, bioaccumulative and toxic chemicals; vPvB refers to very persistent, very bioaccumulative chemicals; POP refers to persistent organic pollutants; PMT refers to persistent, mobile and toxic chemicals; vPvM refers to very persistent, very mobile chemicals; CMR refers to chemicals that are carcinogenic, mutagenic and toxic to reproduction; EDC refers to endocrine disrupting chemicals; STOT refers to chemicals with specific target organ toxicity (after repeated exposure), Aq. Tox. refers to chemicals inducing aquatic toxicity." (Monclús et al. 2025, Supplements p. 13)

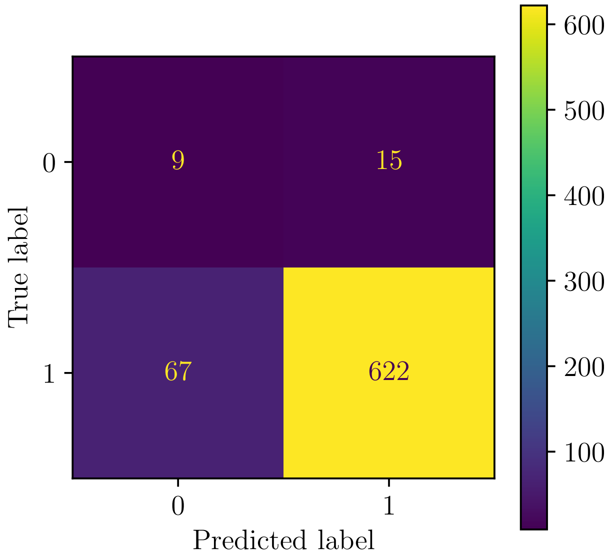

Figure S2: Confusion matrix for the trained model on the independent test set (n=713).

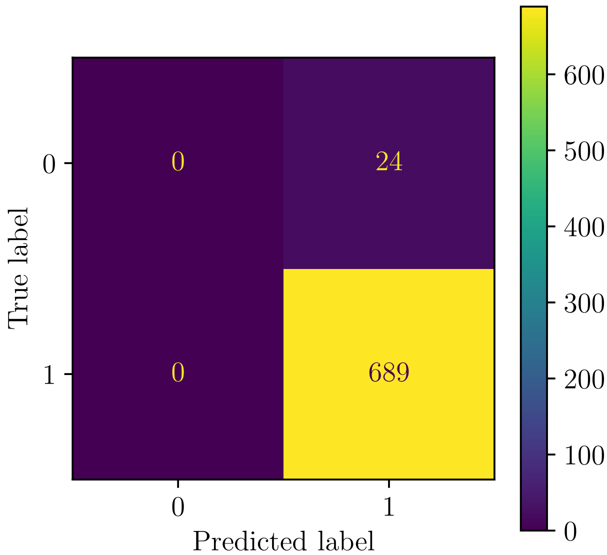

Figure S3: Confusion matrix for the dummy classifier on the independent test set (n=713).

Declaration of Academic Integrity

1. I hereby confirm that this work — or in case of group work, the contribution for which I am responsible and which I have clearly identified as such — is my own work and that I have not used any sources or resources other than those referenced. I take responsibility for the quality of this text and its content and have ensured that all information and arguments provided are substantiated with or supported by appropriate academic sources. I have clearly identified and fully referenced any material such as text passages, thoughts, concepts or graphics that I have directly or indirectly copied from the work of others or my own previous work. Except where stated otherwise by reference or acknowledgement, the work presented is my own in terms of copyright.

2. I understand that this declaration also applies to generative AI tools which cannot be cited (hereinafter referred to as 'generative AI'). I understand that the use of generative AI is not permitted unless the examiner has explicitly authorized its use (Declaration of Permitted Resources). Where the use of generative AI was permitted, I confirm that I have only used it as a resource and that this work is largely my own original work. I take full responsibility for any AI-generated content I included in my work. Where the use of generative AI was permitted to compose this work, I have acknowledged its use in a separate appendix. This appendix includes information about which AI tool was used or a detailed description of how it was used in accordance with the requirements specified in the examiner's Declaration of Permitted Resources. I have read and understood the requirements contained therein and any use of generative AI in this work has been acknowledged accordingly (e.g. type, purpose and scope as well as specific instructions on how to acknowledge its use).

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Jena, August 31, 2026

Amelie Hübner